Results dashboard

Wind farm digital twin - results report

Data: CARE to Compare, Wind Farm A (5 turbines, 22 datasets, 12 recorded faults of 4 types).

FORECASTING (power from weather, unseen real data)

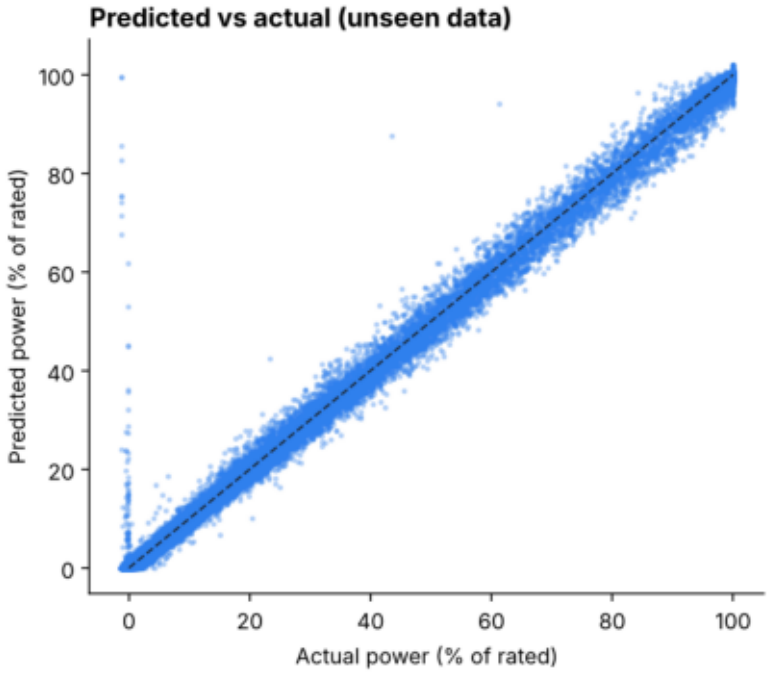
```
  MAE  1.15 % of rated
  RMSE 1.82 % of rated
  R2   0.995
  MAPE 4.8 % (rows above 10 % of rated)
```

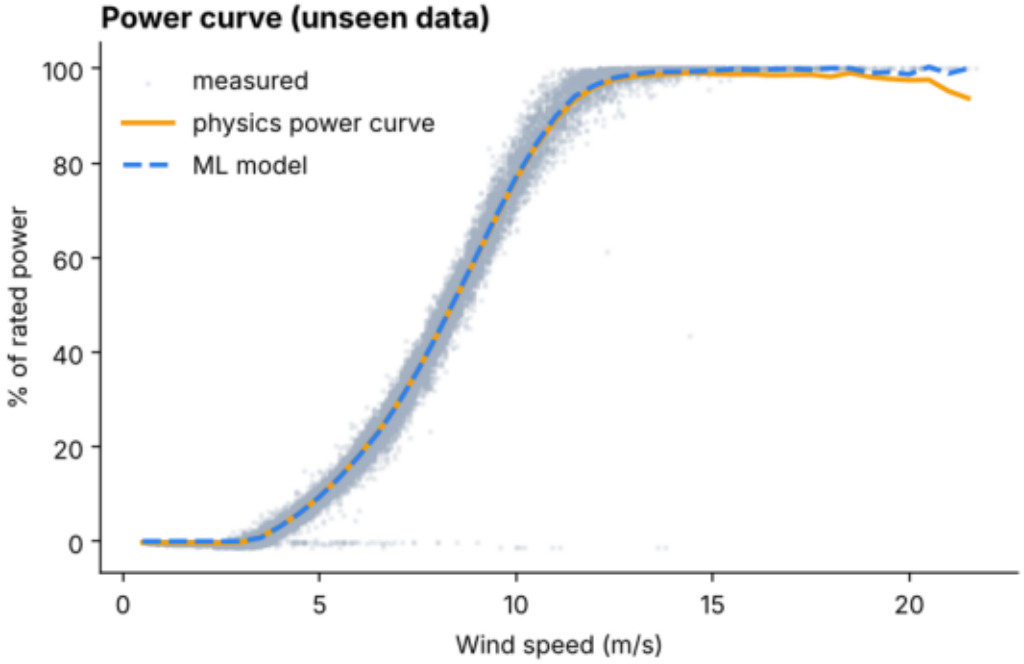
ANOMALY DETECTION (real data, leave-one-turbine-out)

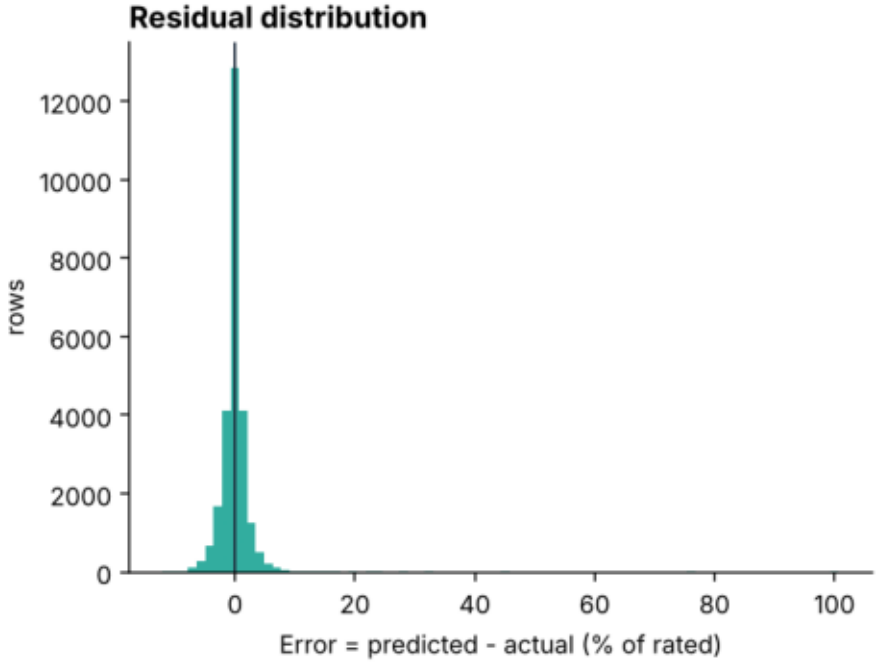
```
  Model in the app: Random Forest (Step 1, approach C), 175 features, class-balanced.
  Cross-check: residual detector (normal-behaviour models) + 24-h energy performance monito
  A  old model (v1 threshold + diagnosis)        detected  1/12  correct  0/12  false alarm
0/10
  B  Gradient Boosting, class-balanced           detected  9/12  correct  1/12  false alarm
7/10
  C  Random Forest, balanced                     detected  8/12  correct  1/12  false alarm
6/10
  D  residual detector + fault-type classifier   detected 10/12  correct  3/12  false alarm
8/10

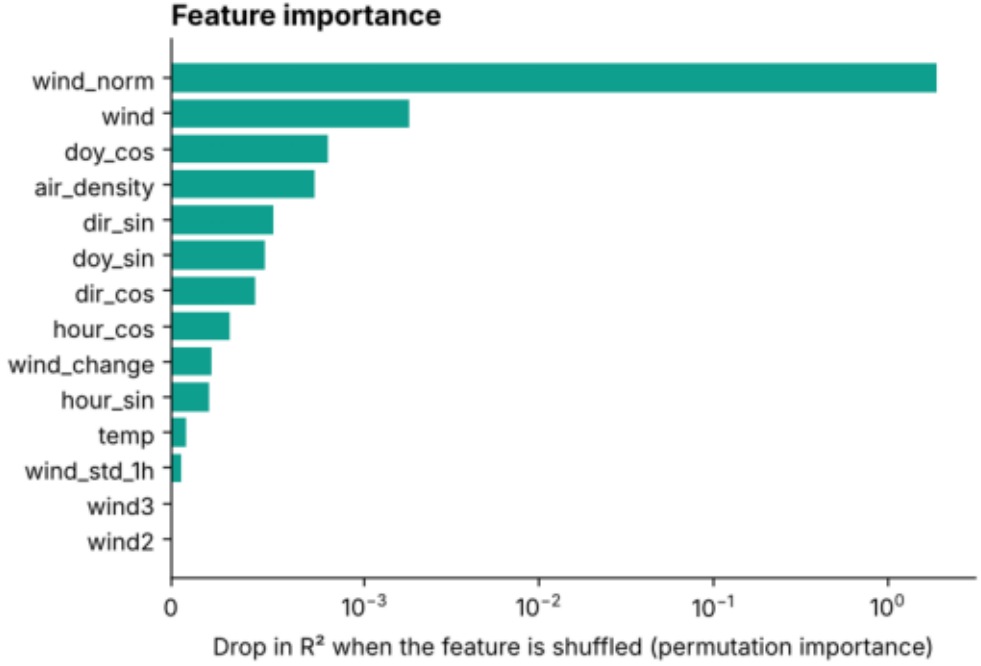
  With 1-6 events per fault type the fault type cannot be classified reliably on unseen
turbines;
  even with settings tuned on all data at most 3 of 12 faults are classified correctly.
```

Forecasting model









Random Forest (C) - per-class metrics (real data, out-of-fold)

class	precision	recall	F1	support (rows)
normal	0.65	0.895	0.753	20357
hydraulic	0.0	0.0	0.0	4387
gearbox	0.0	0.0	0.0	3252
generator_bearing	0.09	0.028	0.043	3450
transformer	0.0	0.0	0.0	829

Confusion matrices (real data)

